

DEPARTMENT OF MATHEMATICS

QUIZ TEST

Sub. Code:	CY35	Sub.:	Discrete Mathematical Structures	Test:	Quiz	Date	17.12.2024
Semester:	III	Section:	CS(AIML)-B	Marks:	10	Time	11.05 to 11.35pm

Note: Answer all the questions, each main question carries 2 marks.

1. Test the following argument for Tautology:
 $[(p \rightarrow q) \wedge \sim p] \rightarrow q$
2. Prove by contradiction method: $\sqrt{2}$ is irrational number.
3. Draw the Hasse diagram representing the partial ordering relation R on the set $A = \{1, 2, 3, 4, 5\}$ such that ' xRy ' if and only if ' $x = y^k$ ' for any positive integer k.
4. Find the explicit formula for the sequence $a_n = a_{n-1} + 2^n$, $a_1 = 3$ using back tracing method.
5. Define Ring sum of two graph with an example.